



UNIVERSITY EXAMINATIONS

2023/2024 ACADEMIC YEAR

REGULAR EXAMINATION
FOR THE DEGREE OF

BACHELOR OF SCIENCE

COURSE CODE: COMP 321

COURSE TITLE: COMPUTER GRAPHICS

DATE: 17TH APRIL 2024

TIME: 2.00 P.M. -5.00 P.M

INSTRUCTIONS TO CANDIDATES

- SEE INSIDE

THIS PAPER CONSISTS OF (3) PRINTED PAGES. PLEASE TURN OVER



UNIVERSITY OF ELDORET

SCHOOL OF SCIENCE

DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE

EXAMINATION FOR THE BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

COMP321: COMPUTER GRAPHICS

INSTRUCTIONS: Answer all questions on section A and any other THREE Questions

Section A: Answer ALL questions**QUESTION ONE (16 MKS)**

- a. Define the term 'Raster Graphics' and explain its advantages and disadvantages. [5 Marks]
- b. Describe the differences between Vector Graphics and Bitmap Graphics. [4 Marks]
- c. Explain the concept of Anti-Aliasing in computer graphics and its significance. [4 Marks]
- d. Discuss the importance of Color Depth in digital images and its impact on image quality. [3 Marks]

QUESTION TWO (15 MKS)

- a. Elaborate on the role of GPUs in computer graphics rendering and acceleration. [7 Marks]
- b. Compare and contrast Ray Tracing and Rasterization techniques in rendering images. [8 Marks]

Section B: Answer any THREE questions**QUESTION THREE (13 Mks)**

- a. Explain the principles of 3D modeling and rendering in computer graphics. [6 Marks]
- b. Discuss the different types of shading models used in computer graphics and their applications. [4 Marks]
- c. Describe the process of Texture Mapping and its significance in enhancing visual realism. [3 Marks]

QUESTION FOUR (13 Mks)

- a. What is the purpose of Z-buffering in 3D graphics rendering? [2 Marks]
- b. Explain the concept of Polygon Meshes and their role in representing 3D objects. [6 Marks]
- c. Discuss the challenges and solutions related to Hidden Surface Removal in computer graphics. [5 Marks]

QUESTION FIVE (13 Mks)

- a. Describe the concept of Animation in computer graphics and its applications. [6 Marks]
- b. Explain the importance of Lighting and Shadows in creating realistic visual effects in computer graphics. [4 Marks]

- c. Discuss the advantages and disadvantages of Procedural Generation techniques in computer graphics. [3 Marks]

QUESTION SIX (13 Mks)

- a. Define the term 'Rendering Pipeline' in computer graphics and outline its stages. [5 Marks]
- b. Discuss the role of Transformations in manipulating objects in a 3D environment. [4 Marks]
- c. Explain the concept of GPU Accelerated Computing and its impact on real-time graphics rendering. [4 Marks]

QUESTION SEVEN (13 Mks)

- a. Differentiate between 2D and 3D computer graphics rendering techniques. [4 Marks]
2D: single viewpoint, multiple angles
- b. Discuss the importance of Frame Buffering in rendering images and videos. [5 Marks]
stores info regarding color depth, shadows, quick access
- c. Explain the concept of Ray Casting and its applications in computer graphics. [4 Marks]
shadows, reflections → realistic, shadows & manipulation of object's luminance.